EPIGENETICS

Module map

- Module 10: Epigenetics
- Connected lab: lab-11_epigenetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 10
Epigenetics

1. Gene regulation as control
2. DNA methylation
3. Histone and chromatin change
4. Environment and expression
5. Epigenetics versus mutation

Lab evidence
lab-11_epigenetics.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

EPIGENETICS

Learning objectives

- Explain gene regulation as control of gene activity.
- Describe DNA methylation at an introductory level.
- Connect chromatin structure to gene accessibility.
- Give a careful example of environmental influence on expression.
- Distinguish epigenetic regulation from mutation.

OBJECTIVE 1
Explain gene regulation as control of gene activity.

OBJECTIVE 2
Describe DNA methylation at an introductory level.

OBJECTIVE 3
Connect chromatin structure to gene accessibility.

OBJECTIVE 4
Give a careful example of environmental influence on expression.

OBJECTIVE 5
Distinguish epigenetic regulation from mutation.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

EPIGENETICS

Topic sequence

- Gene regulation as control: Cells regulate which genes are active rather than using every gene equally.
- DNA methylation: DNA methylation can reduce transcription without changing base sequence.
- Histone and chromatin change: Histone and chromatin changes alter access to DNA.
- Environment and expression: Environmental conditions can influence gene expression patterns.
- Epigenetics versus mutation: Epigenetic change affects gene activity, while mutation changes DNA sequence.

01

Gene regulation as control

Cells regulate which genes are active rather than using every gene equally.

02

DNA methylation

DNA methylation can reduce transcription without changing base sequence.

03

Histone and chromatin change

Histone and chromatin changes alter access to DNA.

04

Environment and expression

Environmental conditions can influence gene expression patterns.

05

Epigenetics versus mutation

Epigenetic change affects gene activity, while mutation changes DNA sequence.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

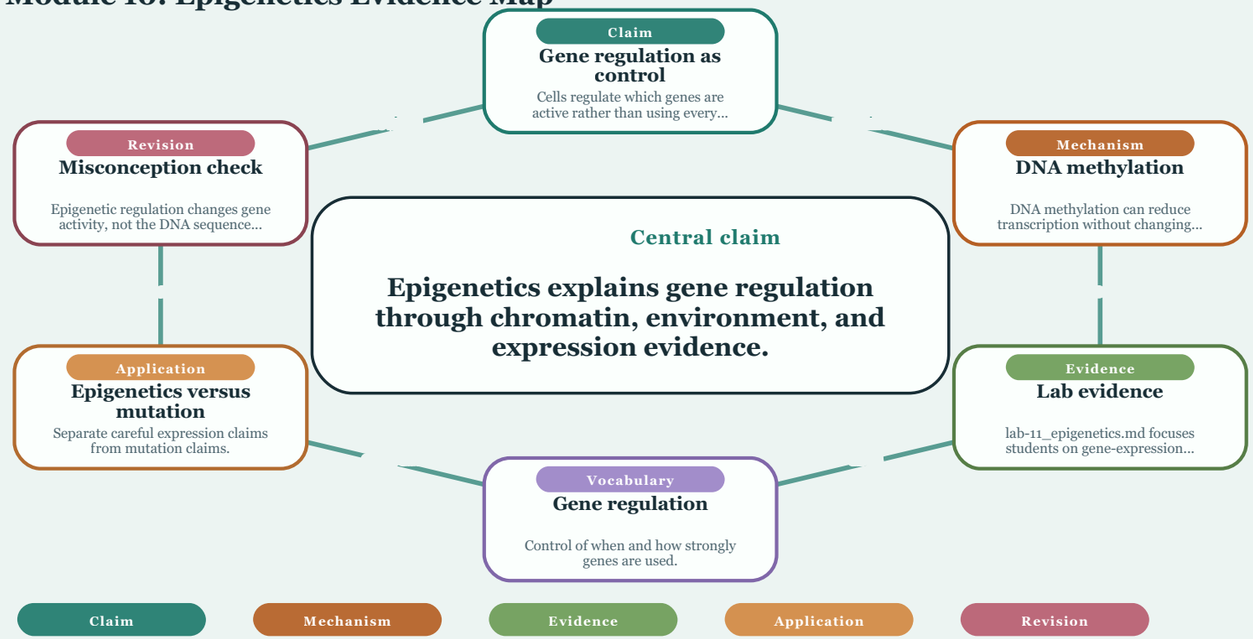
EPIGENETICS

Module 10: Epigenetics Evidence Map

- Module focus: Epigenetics
- Central claim: Epigenetics explains gene regulation through chromatin, environment, and expression evidence.
- Key nodes to connect: Gene regulation as control, DNA methylation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-11_epigenetics.md supplies an evidence surface for the map.

Module 10 / Concept Map

Module 10: Epigenetics Evidence Map



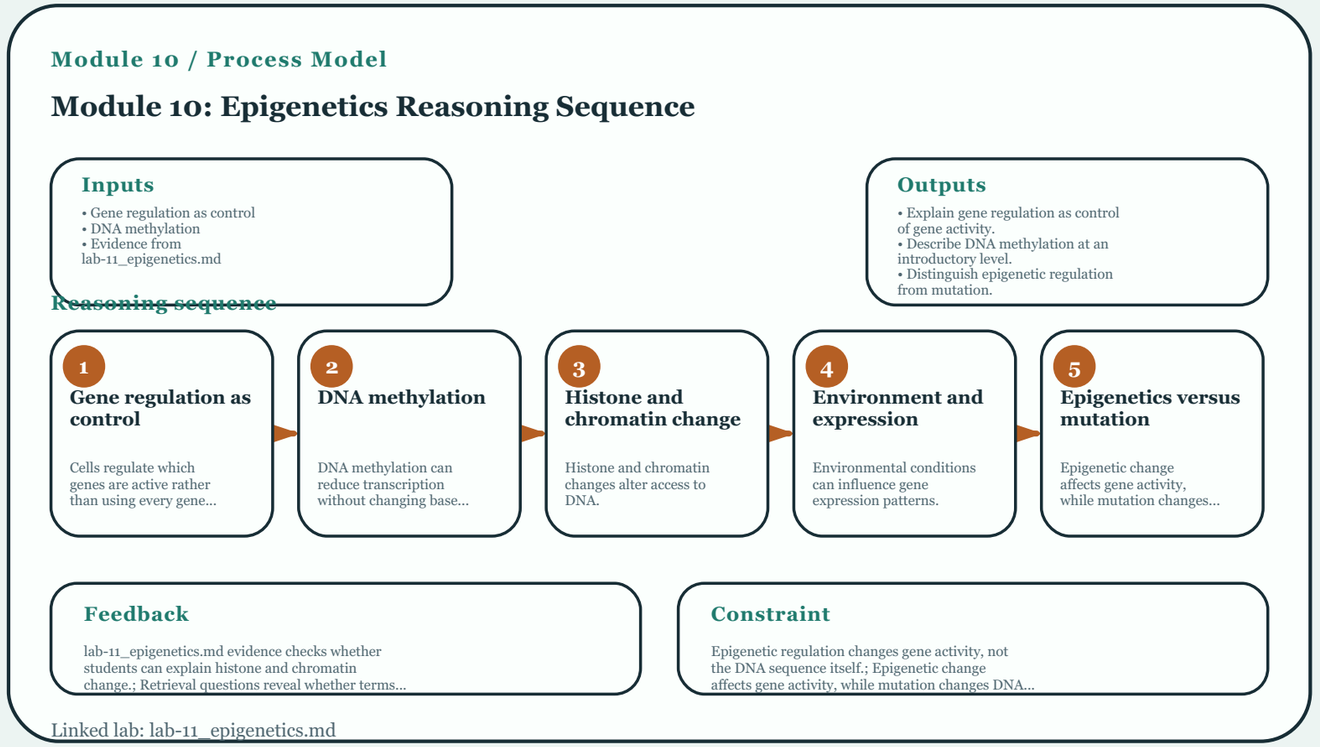
Linked lab: lab-11_epigenetics.md / Generated from explicit module.toml visual schema

Teaching note: Ask students to explain one edge in their own words before moving on.

EPIGENETICS

Module 10: Epigenetics Reasoning Sequence

- Module focus: Epigenetics
- Trace the model: Gene regulation as control -> DNA methylation -> Histone and chromatin change -> Environment and expression
- Inputs become outputs: Gene regulation as control, DNA methylation -> Explain gene regulation as control of gene activity., Describe DNA methylation at an introductory level.
- Feedback check: lab-11_epigenetics.md evidence checks whether students can explain histone and chromatin change.
- Lab connection: lab-11_epigenetics.md gives students a process to observe or test.



Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

EPIGENETICS

Terms as evidence handles

- Gene regulation: Control of when and how strongly genes are used.
- Epigenetics: Heritable or persistent expression changes without DNA sequence change.
- DNA methylation: Chemical marking of DNA often associated with reduced transcription.
- Histone: Protein around which DNA is packaged.
- Chromatin: DNA-protein complex that can be open or compact.
- Transcription factor: Protein that helps control transcription.

GENE REGULATION
Control of when and how strongly genes are used.

EPIGENETICS
Heritable or persistent expression changes without DNA sequence change.

DNA METHYLATION
Chemical marking of DNA often associated with reduced transcription.

HISTONE
Protein around which DNA is packaged.

CHROMATIN
DNA-protein complex that can be open or compact.

TRANSCRIPTION FACTOR
Protein that helps control transcription.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

EPIGENETICS

Lab connection

- Lab file: lab-11_epigenetics.md
- Lab evidence should test or illustrate: Histone and chromatin change
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Gene regulation as control

EVIDENCE
lab-11_epigenetics.md

CLAIM
Explain gene regulation as control of gene activity.

Teaching note: Make the lab connection explicit before students encounter the activity.

EPIGENETICS

Contrast check

- Do not stop at: Cells regulate which genes are active rather than using every gene equally.
- Stronger explanation: Epigenetic change affects gene activity, while mutation changes DNA sequence.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Cells regulate which genes are active rather than using every gene equally.

DEEPER

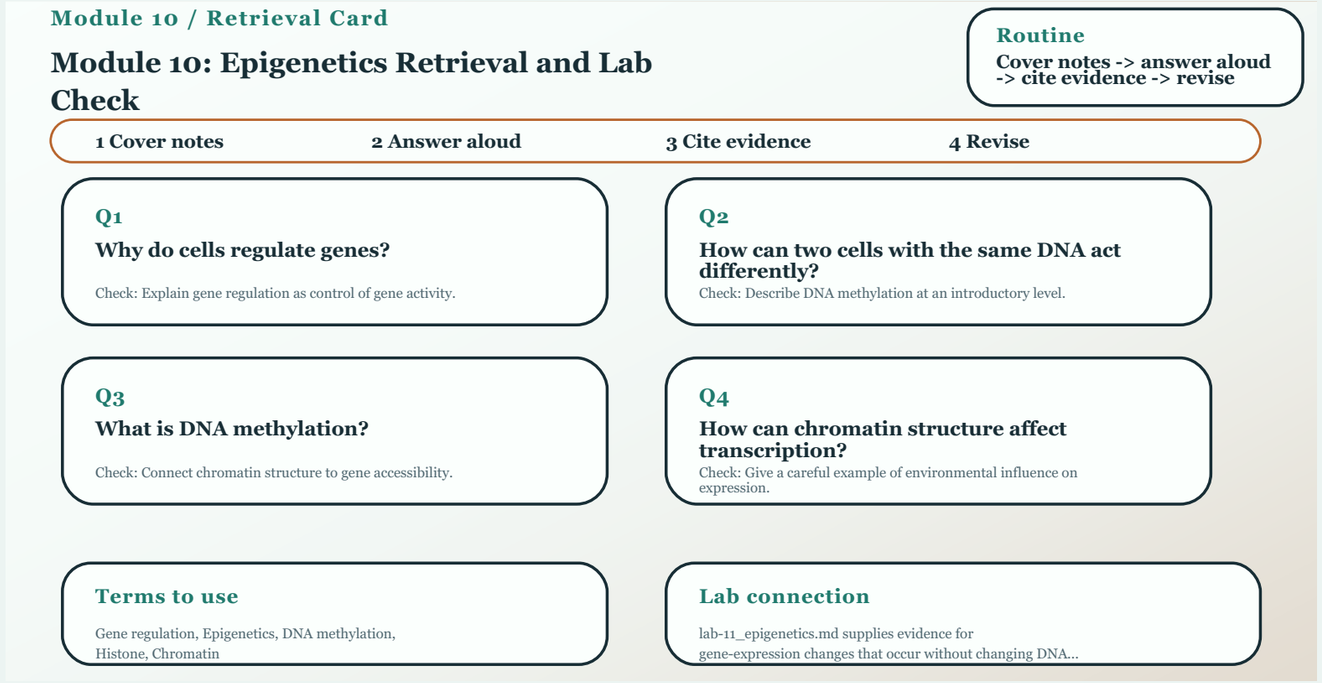
Epigenetic change affects gene activity, while mutation changes DNA sequence.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

EPIGENETICS

Module 10: Epigenetics Retrieval and Lab Check

- Module focus: Epigenetics
- Retrieval prompt: Why do cells regulate genes?
- Answer check: Explain gene regulation as control of gene activity.
- Required terms: Gene regulation, Epigenetics, DNA methylation
- Lab connection: lab-11_epigenetics.md; lab-11_epigenetics.md supplies evidence for gene-expression changes that occur without changing DNA sequence.



Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

EPIGENETICS

Practice quiz bridge

- 1. How does epigenetic regulation differ from a mutation?
- 2. DNA methylation is usually associated with:
- 3. When chromatin is packed tightly, genes in that region are generally:
- 4. Two genetically identical twins come to differ in a trait over their lifetimes. A good explanation is that:

QUESTION 1

Key: C

How does epigenetic regulation differ from a mutation?

QUESTION 2

Key: A

DNA methylation is usually associated with:

QUESTION 3

Key: D

When chromatin is packed tightly, genes in that region are generally:

QUESTION 4

Key: B

Two genetically identical twins come to differ in a trait over their lifetimes. A good explanation is that:

Teaching note: Use the quiz as formative feedback. Answer key: 1=C, 2=A, 3=D, 4=B

EPIGENETICS

Synthesis and exit ticket

- Exit claim: explain gene regulation as control using evidence from lab-11_epigenetics.md.
- Must include: Gene regulation, Epigenetics, DNA methylation.
- Revision prompt: what evidence would change your explanation?

CLAIM
Gene regulation as control

EVIDENCE
lab-11_epigenetics.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.